

SKYGUARD AI

Edge-AI Weather QC Transmit-Only-on-Suspicion

72% energy saving • ~10-month battery • self-healing AWS network

Problem Statement: AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)

Theme: Smart Automation & Observational Networks | **Category:** Software (with Edge-AI hardware)

Parameters: Temperature °C • Atmospheric Pressure hPa • Relative Humidity %

Team ID: _____ | **Team Name:** _____ | **Mentor:** _____

The Insight

Most AWS faults are catchable on-device with a tiny model. Why radio every reading upstream when ~70-80% of readings are clean? **Transmit only on suspicion** → massive energy + bandwidth saving.

Edge + Cloud Split



On-ESP32 (catches ~70-80% locally)

- L1 Physics (range, dew-point, barometric, rate) — O(1) arithmetic
- L3 Frozen (std < 0.01) — O(window)
- Gap/dropout detection
- **int8 Isolation Forest** via emlearn — <10 KB flash, <1 KB RAM, <10ms inference
- **int8 MLP autoencoder** via TFLite-Micro — 5-50 KB flash, 30-80 KB arena, 1-10ms (ESP32-S3 with ESP-NN)

On Gateway/Cloud (only suspect readings)

- L2 full ML (Ext. IsoForest + SHAP + LSTM-AE + trend)
- L4 Chronos-2 (120M, multivariate) corridor breach
- L5 Spatial robust-Z (needs network-wide neighbour state)
- L6 Fusion 2x2 matrix

The Fusion Matrix

	Neighbours AGREE (Z < 3.1)	Neighbours DISAGREE (Z ≥ 3.1)
Model flagged (L1-L4)	REGIONAL_EVENT real weather — alert, not fault — Medium	SENSOR_FAULT sensor is broken — High
No model flag	HEALTHY all clear — no action	CHECK_HEALTH subtle drift — schedule maintenance — Low

Novelty

- **No competitor does per-station edge ML** (Vaisala embeds in firmware; WeatherXM/Tomorrow.io always-transmit). IEEE 2026 did edge-IF on Raspberry Pi, not ESP32.
- **Transmit-only-on-suspicion** is a unique energy/deployability story.
- ESP32-S3 ≈ ₹500 vs industrial AWS → **democratizes weather QC.**

ESP32-S3 Energy Budget (measured/datasheet)

Mode	Current	Source
Deep sleep (module)	8 μ A	ESP32-S3 datasheet
WiFi TX 20.5 dBm	355 mA peak	ESP32-S3 WROOM-1
LoRa TX 17 dBm	330 mA	Heltec V4 datasheet
LoRa TX 27 dBm	750 mA	Heltec V4 datasheet
Sensor sample + edge ML	$\sim$ 50 mA $\times$ 200 ms	estimated

Daily Energy (5-min sampling, 288/day)

Strategy	mAh/day
Always-transmit (LoRa @17dBm)	$\sim$ 34.6
Anomaly-only transmit (5% rate)	$\sim$9.6
Saving	$\sim$72%

Battery life (3000 mAh): $\sim$ 10 months anomaly-only vs $\sim$ 3 months always-transmit. Source: ESP32-S3 + Heltec V4 datasheets.

Stack

Tier	Technology
Edge	ESP32-S3 + emlearn + TFLite-Micro + LoRa/Wi-Fi
Gateway	Node.js + Fastify + WebSocket
ML core	Python + FastAPI; EIF, SHAP, LSTM-AE, chronos-2, alibi-detect
Store/Cache	TimescaleDB + Redis + BullMQ
Frontend	React + Leaflet + Recharts

Key Thresholds (sourced)

Check	Threshold	Source
Temp rate (fault)	$ \Delta T > 19.4\text{ }^{\circ}\text{C/hr}$	NOAA MADIS (35 $^{\circ}$ F/hr)
Temp rate (alert)	$ \Delta T > 8\text{ }^{\circ}\text{C/hr}$	Tightened watch level
Pressure rate (fault)	$ \Delta P > 15\text{ hPa/hr}$	NOAA MADIS sea-level
Pressure rate (alert)	$ \Delta P > 3\text{ hPa/hr}$	Cyclone-class (weather, not fault)
Humidity rate (fault)	$ \Delta RH > 50\text{ \%/hr}$	NOAA MADIS
Dew-point constraint	$T_{\text{dew}} \leq T_{\text{air}} + 0.5^{\circ}\text{C}$	Magnus / Alduchov-Eskridge; MADIS
Barometric altitude	$ P - P_{\text{exp}} > 30\text{ hPa}$	Barometric formula
Range bounds (India)	$T\text{ }-40/+55^{\circ}\text{C}$, RH 0–100%, P 300–1084	WMO records, BoM
Climatological	monthly mean $\pm 5\sigma$	WMO guideline / Lanzante 1996
Spatial robust-Z	$Z \geq 3.1 \rightarrow$ disagree	Met Norway TITAN default
Frozen sensor	std < 0.01 over ≥ 10 readings	OK Mesonet persistence test
Forecast breach	actual $\notin$ [p5,p95] or severity > 0.8	Chronos-2 corridor
Drift (Mann-Kendall)	$p < 0.05$ + Theil-Sen slope	Standard trend test

Practical Deployability

- **Cost:** ESP32-S3 ≈ ₹500/station vs industrial AWS (₹lakhs).
- **Standards-aligned:** WMO L0-L5 (CIMO Guide WMO-No. 8; MADIS; WQMS).
- **Free data:** Open-Meteo (ERA5, no key) + NOAA ISD (public domain).
- **Drop-in:** works with existing IMD AWS API for production ingest.
- **Components validated:** edge-IF IEEE 2026 (<10ms/<200KB); Met Éann LSTM-AE 99.6%; ECMWF ML operational.
- **Scale path:** 15 demo → 675 IMD → 20,000+ NOAA; K8s + BullMQ + TimescaleDB.
- **Security:** HMAC-signed telemetry + tamper classifier (closes a verified WMO gap — no standard mandates signed AWS data).

Risks & Mitigations

Risk	Mitigation
ESP32 PSRAM latency	int8 quant; weights in flash not PSRAM where possible
OTA corruption	ESP-IDF rollback partition; fail-safe boot
ADC noise ±7%	Calibration + moving-average filter pre-ML
Radio TX peak current	Transmit-only-on-suspicion → ~5% duty cycle; solar + Li-ion buffer
IsoForest blind spots	Extended IF + LSTM-AE sibling + PSI retrain
ML flags real weather	Spatial consensus → REGIONAL_EVENT; de-seasonalize
Isolated stations	Multi-layer agreement + NWP virtual buddy
Regulatory / black-box	SHAP + human-in-loop + deterministic fallback + audit log

Sustainability

72% energy

~10 mo battery

Less e-waste

Fewer field trips

Target Audience

- IMD & State met depts (675+ AWS)
- Aviation, agriculture, disaster mgmt
- Climate research, NWP assimilation
- Low-cost / citizen weather networks (democratization)

Scale

675+ IMD

20,000+ NOAA

₹500/station

Benefits

Type	Benefit
Social	Better disaster warnings → lives saved
Economic	₹500 vs ₹lakhs hardware; predictive maintenance; 72% less energy
Environmental	~10 mo battery; less e-waste; fewer truck rolls
Operational	Fix sensors before bad data flows; self-healing imputation

User Experience — Explained Alerts

Alert card = verdict + confidence + blamed sensor (SHAP) + reason + action + imputed value.

Dashboard

- Leaflet map — station colour = health
- Live alert feed — severity colour-coded
- SHAP bar chart per flagged reading
- Chronos-2 forecast corridor chart
- Sensor-health panel + predicted time-to-failure

Expected Outputs (problem statement)

Real-time alerts • severity + confidence • root-cause class • dashboard • sensor health • corrected-data estimation — **all delivered**.

Self-Healing Loop

Degraded sensor → auto-upweight neighbour/Chronos-imputed values downstream → **trustworthy data flows uninterrupted** → the grand-challenge answer.

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